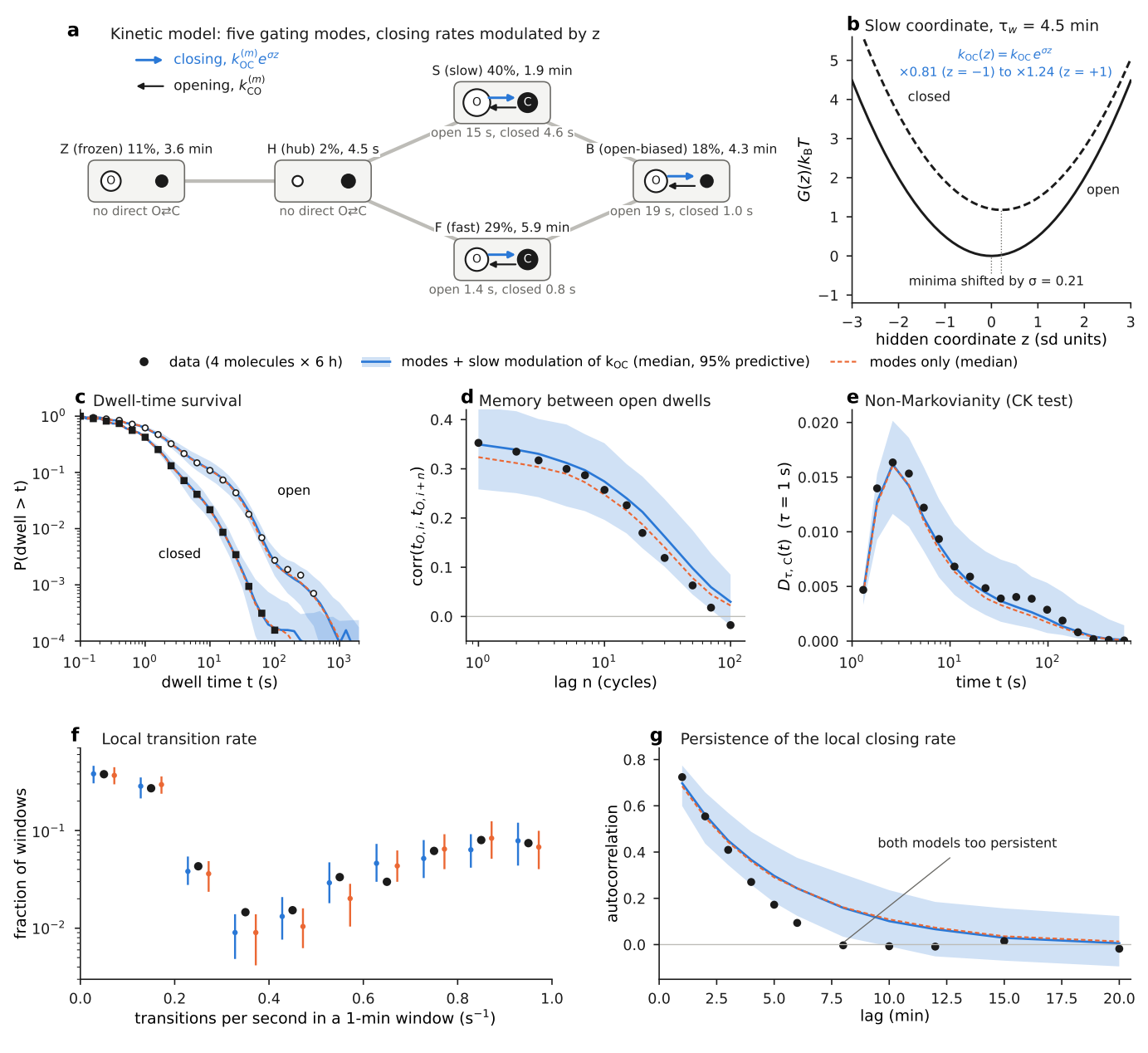
a
Kinetic model: five gating modes, closing rates modulated by z
closing, $k_{OC}^{(m)} e^{\alpha z}$
opening, $k_{CO}^{(m)}$
S (slow) 40%, 1.9 min
open 15 s, closed 4.6 s
Z (frozen) 11%, 3.6 min
no direct O⇄C
H (hub) 2%, 4.5 s
no direct O⇄C
B (open-biased) 18%, 4.3 min
open 19 s, closed 1.0 s
F (fast) 29%, 5.9 min
open 1.4 s, closed 0.8 s
b
Slow coordinate, τ_w = 4.5 min
$k_{OC}(z) = k_{OC} e^{\alpha z}$
×0.81 (z = −1) to ×1.24 (z = +1)
closed
open
minima shifted by σ = 0.21
$G(z)/k_BT$
hidden coordinate z (sd units)
data (4 molecules × 6 h)
modes + slow modulation of k_{OC} (median, 95% predictive)
modes only (median)
c
Dwell-time survival
P(dwell > t)
open
closed
dwell time t (s)
d
Memory between open dwells
corr($t_{O,i}$, $t_{O,i+n}$)
lag n (cycles)
e
Non-Markovianity (CK test)
$D_{\tau,C}(t)$ (τ = 1 s)
time t (s)
f
Local transition rate
fraction of windows
transitions per second in a 1-min window (s^{-1})
g
Persistence of the local closing rate
autocorrelation
both models too persistent
lag (min)